Chapter 6: Why hypothesis testing?



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You are accused of a murder… When will you be labelled as "guilty"?

- In a civilian court:
  - It must be proven that you did commit the murder before you are labelled as guilty.
  - Underlying assumption: "You are NOT GUILTY".
  - The judge has to decide your innocence or guilt on the basis of incomplete evidence.

In statistics terminology:

Null hypothesis → H_0 : You are NOT GUILTY

Alternative hypothesis → H_1 : You are GUILTY

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You are accused of a murder… When will you be labelled as "guilty"?

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What are the possible errors and their consequences?

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Analysis of hypotheses testing

Two possible conclusions from hypotheses-testing analysis:
- reject H_0
- fail to reject H_0

	The truth about the murder	
Based upon the sample evidence judge decides to	H_0: True (You are not guilty)	H_0: False (You are guilty)
Do not reject H_0	✓	Type-II
Reject H_0	Type –I	✓

What are the social costs of Type-1 and Type-2 error?

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Analysis of hypotheses testing

The null hypothesis H_0 : conservative

- Type-1 error: reject H_0 when H_0 is true.
- P(reject H_0 when H_0 is true) is controlled directly.

Rejecting H_0 is a strong conclusion.

The alternative hypothesis H_a :

- Type-2 error: do not reject H_0 when H_a is true.
- P(do not reject H_0 when H_0 is not true) depends on the true value of the parameter and the sample size.

Accepting H_0 is a weak conclusion. → Fail to reject H_0 *or* Cannot reject H_0 *or* Do not reject H_0

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Are Maine's lakes above the safety limits?

Federal law: The mean mercury level should be below 1.0 ppm

State law: The mean mercury level should be below 0.43 ppm

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If you were the decision maker, how would you go about "issuing the health advisory??"

- Formulate a hypothesis testing to decide whether there is enough evidence for the Maine fish to be poisonous.
- Two possible actions:
  - To issue a health advisory. ← Result of the hypothesis that the Maine fish are poisonous.
  - Not to issue a health advisory. → Result of the hypothesis that the Maine fish are not poisonous.

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If the state of Maine is very concerned about the health of the citizens, how would you choose hypotheses of "the fish"?

- Conservative hypothesis:
  The Maine fish are poisonous.
- How to state this hypothesis in statistics?
  - $\mu = 0.43$ppm
- What is the contradictory hypothesis?
  - $\mu < 0.43$ppm

$H_0: \mu = 0.43$
$H_a: \mu < 0.43$

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If the state of Maine is very concerned about the wealth of the fisheries, or it prefers to keep the status quo, how would you choose hypotheses of "the fish"?

- Conservative hypothesis:

  The Maine fish are NOT poisonous.

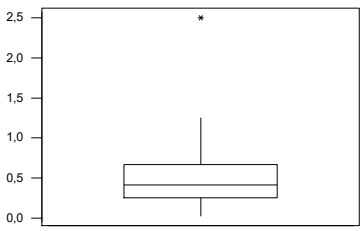
- How to state this hypothesis in statistics?
  - $\mu = 0.43$ppm
- What is the contradictory hypothesis?
  - $\mu > 0.43$ppm

$H_0: \mu = 0.43$
$H_a: \mu > 0.43$

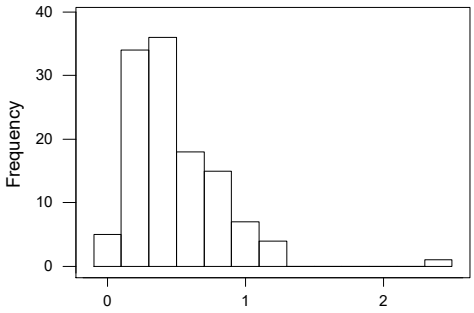
Descriptive statistics and boxplot

Variable	N	Mean	Median	StDev
Mercury level	120	0,4850	0,4100	0,3333

Variable	Minimum	Maximum	Q1	Q3
Mercury level	0,0250	2,5000	0,2500	0,6675



Histogram and skewness



- sample skewness with the outlier: 2.208
- sample skewness without the outlier: 0.741

What to do with the extreme outlier?

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Statistical hypotheses

A statistical hypothesis or hypothesis is a claim about the value of a single population characteristic or several population characteristics.

Definition: The null hypothesis denoted by H_0, is the claim about one or more population characteristics that is initially assumed to be true. The alternative hypothesis, H_1 or H_a, is the assertion contradictory to H_0.

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Example: Milemaster tires

Strategy of Milemaster company: To introduce a new tire only if the true average life of the proposed tire is greater than 47,000 miles.

50 sample tires are produced.

Through accelerated life tests:

$$\bar{x} = 47{,}351 \text{ miles} \quad s = 1924.56 \text{ miles}$$

It is believed that X_i's are normal random variables.

Question: To market or not to market this new tire

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Example: Milemaster tires

μ : true average life of the proposed tire

Null hypothesis

H_o: $\mu \leq 47000$

→ Null action
Do not market the tire

Alternative hypothesis

H_a: $\mu > 47000$

→ Alternative action
Market the tire

H_0 is conservative.

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Example: Milemaster tires

	The truth about the population mean tire life	
Based upon the sample evidence you decide to	H_0: True (47,000 miles or under)	H_0: False (over 47,000 miles)
Do not reject H_0	Correct decision	Type II error
Reject H_0	Type I error	Correct decision

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Example: Milemaster tires

Consequences of errors??

Type-I error: Introduce the tire even if $\mu \leq 47{,}000$.

Very costly

Type-II error: Do not introduce the tire even if $\mu > 47{,}000$.

Depends on the real value of μ:
If $\mu = 47{,}500$: low cost
If $\mu = 57{,}000$: high cost of opportunity

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Example: Milemaster tires

How to decide which hypothesis to believe?

H_o: $\mu \le 47{,}000$ → H_o: $\mu = 47{,}000$

H_a: $\mu > 47{,}000$ H_a: $\mu > 47{,}000$

Reject H_0 if $\bar{X}$ is large enough.

What is large enough?

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Example: Milemaster tires

Type I is serious, so we can set a bound on probability of type-I error.
Using this bound we can find the value of "large enough" to reject H_0.

α : significance level

$$\alpha = \mathrm{P}\{\text{Type I}\} = \mathrm{P}\{\text{reject } \mathrm{H}_0 \text{ when } \mathrm{H}_0 \text{ is true}\}$$

$$= P\{\bar{X} > k, \mu = 47000\} = P\{\frac{\bar{X} - \mu}{S/\sqrt{n}} > \frac{k - \mu_0}{S/\sqrt{n}}\}$$

$$= P\{Z > \frac{k - \mu_0}{S/\sqrt{n}}\} = P\{Z > \frac{k - 47000}{1924.56/\sqrt{50}}\} = P\{Z > z_\alpha\}$$

Let α = 0.05. Do we reject the null hypothesis for $\bar{x} = 47{,}351$ miles?

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